EXAM PAPERS PRACTICE

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Detailed mark scheme

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AQA A Level Further Mathematics (7367)

Topic C: Matrices

Sub topic C4: Invariant Points and Lines

Mark Scheme

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AQA A Level Further Mathematics (7367)

Mark Scheme

Topic	**C: Matrices**
Sub topic	**C4: Invariant Points and Lines**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Obtains two equations in x and y. May be seen as a single vector equation. At least one equation must be correct. Accept a pair of letters other than x and y. Ignore any subsequent incorrect working.	AO1.1a	M1	$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ $\begin{bmatrix} 2x + 3y \\ x + 4y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$ $2x + 3y = x$ and $x + 4y = y$
Obtains any two correct invariant points, with no incorrect points. Condone correct points given as position vectors. NMS: Correct answer scores 2/2. NMS: One correct invariant point and only one incorrect point scores SC1.	AO1.1b	A1	$x = -3y$ $\therefore$ two invariant points are (0, 0) and (−3, 1)
			Total 2 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Commences a proof by correctly setting up an equation using the definition of an invariant point	AO2.1	R1	For an invariant point $\mathbf{M}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$
Pre-multiplies by $\mathbf{M}^{-1}$	AO2.1	R1	Pre-multiply both sides by $\mathbf{M}^{-1}$ $\mathbf{M}^{-1}\mathbf{M}\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{M}^{-1}\begin{pmatrix} x \\ y \end{pmatrix}$
Uses $\mathbf{M}^{-1}\mathbf{M} = \boldsymbol{I}$ and concludes their rigorous mathematical argument to deduce that (x, y) is invariant under S **AG**	AO2.2a	R1	$\mathbf{M}^{-1}\mathbf{M} = \boldsymbol{I}$ hence $\begin{pmatrix} x \\ y \end{pmatrix} = \mathbf{M}^{-1}\begin{pmatrix} x \\ y \end{pmatrix}$ Therefore $\begin{pmatrix} x \\ y \end{pmatrix}$ is invariant under S.

Total 3 marks

Q3.

(a) (i) Reflection

Reflection stated for M1

M1

In (the plane) $z = 0$ (or in the $x - y$ plane)

Either version for A1

A1

2

(ii) Rotation

Rotation stated.

M1

About the y-axis

y-axis

A1

through $\frac{\pi}{3}$ radians

(or 60°)

B1

3

(b) $T_1 T_2 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{-\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 & \frac{-\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{-\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{bmatrix}$

M1 correct order of matrices

A1 fully correct

[N.B $\begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ \frac{\sqrt{3}}{2} & 0 & -\frac{1}{2} \end{bmatrix}$ *scores M0A0]*

M1A1

2

(c) (i) For line of invariants points

$$\begin{bmatrix} k & 2 & -1 \\ 1 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

Set up equations − uses $\mathbf{M}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

M1

$\Rightarrow kx + 2y - z = x \Rightarrow (k-1)x + 2y - z = 0$ ①

$x + y + z = y \Rightarrow x + z = 0$ ②

Two equations correct

A1

$3x + 4y + z = z \Rightarrow 3x + 4y = 0$ ③

All three equations correct

A1

From ② $z = -x$

From ③ $y = \frac{-3x}{4}$

Defines variables in terms of one letter or 2 components in $\begin{bmatrix} 4 \\ -3 \\ -4 \end{bmatrix}$ *correct*

M1

Substitute in ① $(k-1)x - \frac{3}{2}x + x = 0$

$x\left[k - 1 - \frac{3}{2} + 1\right] = 0$

$x\left[k - \frac{3}{2}\right] = 0$

Substitution into other equations or $\begin{bmatrix} 4 \\ -3 \\ -4 \end{bmatrix}$ *correct*

A1

$x \neq 0 \quad \Rightarrow \quad k = \frac{3}{2}$

k -value obtained.

A1

(ii) Line $x = \frac{-4}{3}y = -z$

$$\frac{x}{4} = \frac{y}{-3} = \frac{z}{-4}$$

Or equivalent

B1cao

7

Alternative:

$$\begin{bmatrix} k-1 & 2 & -1 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix}$$

Substitute and set up

(M1)

$r_1 \rightarrow r_1 + r_2$

Row operation

(A1)

$$\begin{bmatrix} k & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix}$$

$r_1 \rightarrow r_1 - \frac{1}{2}\ r_3$

Row operation

(A1)

$$\begin{bmatrix} k-\frac{3}{2} & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 3 & 4 & 0 & 0 \end{bmatrix} \Rightarrow k = \frac{3}{2}$$

k-value obtained

(A1)

$$v = \lambda \begin{pmatrix} 4 \\ -3 \\ -4 \end{pmatrix}$$

M1 obtains v in terms of single vector.
A1 correct

(M1A1)

$$\frac{x}{4} = \frac{y}{-3} = \frac{z}{-4}$$

Correct form

(B1cao)

(7)

[14]

Q4.

(a) Determinant of matrix = −8 + 9 = 1

Finding determinant and multiplying by area

M1

Area = 3 × 1 = 3 (square units)

CAO − must show multiplication or refer to scale factor / invariant area or equivalent

A1

2

(b) (i) $\begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix}\begin{bmatrix} x \\ mx+c \end{bmatrix} = \begin{bmatrix} x' \\ y' \end{bmatrix}$

⇒

$(x') = 4x + 3(mx + c)$

$(y') = -3x - 2(mx + c)$

x', y' in terms of x, y, m, c

M1

Invariant lines ⇒ $y' = mx' + c$

⇒ $-3x - 2mx - 2c = 4mx + 3m^2x + 3mc + c$

⇒ $0 = (3m^2 + 6m + 3)x + 3mc + 3c$

Use of $y' = mx' + c$

A1

⇒ $3m^2 + 6m + 3 = 0$ $\quad 3mc + 3c = 0$

Attempt at solving equations where coefficients = 0 or compares coefficients

M1

$3(m + 1)^2 = 0 \qquad 3c(m + 1) = 0$

⇒ $m = -1 \qquad c$ can be any value

Finding the correct value of m

A1

⇒ lines are $y = -x + c$

Fully correct line − no restriction on c

A1

5

SPECIAL CASES

$$\begin{pmatrix} 4 & 3 \\ -3 & -2 \end{pmatrix}\begin{pmatrix} x \\ -x+c \end{pmatrix} = \begin{pmatrix} x+3c \\ -x-2c \end{pmatrix}$$

$x' = x + 3c$

$y' = -x - 2c$

SC1 − Correct multiplication as shown

Consider

$-x' + c$
$= -(x + 3c) + c$
$= -x - 3c + c$
$= -x - 2c$
$= y'$

Hence $y = -x + c$ is an invariant line

SC2 − correct multiplication as shown above and full algebraic solution using $y' = -x' + c$

(ii) When $c = 0$, $y = -x$ is a line of invariant points

Any equivalent form

B1

1

[8]

Q5.

(a) Char. Eqn. is $\lambda^2 - 8\lambda - 9 = 0$

Attempted

M1

Quadratic solved to get two roots

dM1

$\Rightarrow \lambda = 9, -1$

A1

Subst^g. back λ (at least once):

M1

$\lambda = 9 \Rightarrow -x + y = 0 \Rightarrow \lambda = 9$ has evecs. $\alpha \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

any $\alpha \neq 0$

$\lambda = -1 \Rightarrow x + y = 0 \Rightarrow \lambda = -1$ has evecs. $\beta \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

any $\beta \neq 0$

A1

6

(b) (0, 0)

B1

1

[7]

Q6.

(a) Attempt at Char.Eqn.

M1

$\lambda^3 - 9\lambda^2 + 24\lambda - 16 = 0$

One each following coefft.

A3,2,1

Attempt at (at least) one linear factor

M1

$(\lambda - 1)(\lambda - 4)^2 = 0$

A1

$\lambda = 1, 4, 4$

A1

7

(b) $\lambda = 1 \Rightarrow \begin{matrix} 2x - y + z = 0 \\ -x + 2y + z = 0 \\ x + y + 2z = 0 \end{matrix} \Rightarrow \alpha \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

Subst^g. back and solving

M1

A1

$\lambda = 4 \Rightarrow x + y - z = 0$

B1

Choosing any two independent vectors satisfying this

M1

E.g.s: $\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\2\end{bmatrix}$

A1

5

(c) $\alpha\begin{bmatrix}1\\1\\-1\end{bmatrix} \equiv$ a line of invariant points

Invariant line

M1

LoIPs

A1

e.g. $\alpha\begin{bmatrix}1\\0\\1\end{bmatrix} + \beta\begin{bmatrix}0\\1\\1\end{bmatrix} \equiv$ an invariant plane

B1

3

[15]

Q7.

(a) (i) $R_1' = R_1 + R_2$ or $R_2' = R_2 + R_1$

M1

leading to $R_{1/2} = (4 - q \quad 4 - q \quad 0)$

A1

2

(ii) $$\Delta = (4-q)\begin{vmatrix}1 & 1 & 0\\-12 & -1-q & -7\\6 & 6 & 10-q\end{vmatrix}$$

$$= (4-q)\begin{vmatrix}0 & 1 & 0\\q-11 & -1-q & -7\\0 & 6 & 10-q\end{vmatrix}$$

By $C_1' = C_1 - C_2$ (eg)

M1

$$= (4-q)(q-11)\begin{vmatrix}0 & 1 & 0\\1 & -1-q & -7\\0 & 6 & 10-q\end{vmatrix}$$

2^{nd} linear factor correct

A1

$= (4 - q)(q - 11) \times \ldots$

Full attempt at 3^{rd} factor

M1

$= (4 - q)(q - 11)(q - 10)$

A1

4

(b) (i) $\begin{bmatrix} 16 & 5 & 7 \\ -12 & -1 & -7 \\ 6 & 6 & 10 \end{bmatrix}\begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix} = \begin{bmatrix} 8 \\ 20 \\ -28 \end{bmatrix}$

M1A1

$= 4\begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix}$ so that $\lambda = 4$

A1

3

(ii) $\lambda = 10, 11$ noted or used

ft

B1

$$\lambda = 10 \Rightarrow \begin{aligned} 0 &= 6x + 5y + 7z \\ 0 &= -12x - 11y - 7z \\ 0 &= 6x + 6y \end{aligned}$$

B1

Using $y = -x$ to get $z \sim x/y$

M1

Eigenvector $(-7, 7, 1)$

Any non-zero multiple will do

A1

$$\lambda = 11 \Rightarrow \begin{aligned} 0 &= 5x + 5y + 7z \\ 0 &= -12x - 12y - 7z \\ 0 &= 6x + 6y - z \end{aligned}$$

Either $y = -x$ or $z = 0$

M1

Eigenvector (1, –1, 0)

Any non-zero multiple will do

A1 7

(c) $\frac{x}{2} = \frac{y}{5} = \frac{z}{-7}$, $\frac{x}{-7} = \frac{y}{7} = z$

Any line eqns.

M1

or $x = -y$, $z = 0$

any one ft correct

A1 2 **[18]**

Q8.

(a) (i) Evals $\lambda = 27$ 1

Both

B1

Evecs $(\alpha)\begin{bmatrix}4\\1\end{bmatrix}$ and $(\beta)\begin{bmatrix}-1\\3\end{bmatrix}$

Correctly matched up with evals
(look out for λ_1, $\mathbf{v}_1$ notations)

B1B1B1 4

(ii) $y = -3x$

Ft 4y = x if evecs mis-matched

B1ft

from $\lambda = 1$

***Must** say why they have chosen this one*

B1 2

(b) $\mathbf{U}^{-1} = \frac{1}{13}\begin{bmatrix}3 & 1\\-1 & 4\end{bmatrix}$

Det; mtx

B1B1

M = **U D U**$^{-1}$

*Including attempt to multiply (at least **U D** ...)*

M1

$= \frac{1}{13}\begin{bmatrix}4 & -1\\1 & 3\end{bmatrix}\begin{bmatrix}27 & 0\\0 & 1\end{bmatrix}\begin{bmatrix}3 & 1\\-1 & 4\end{bmatrix}$

$= \frac{1}{13}\begin{bmatrix}4 & -1\\1 & 3\end{bmatrix}\begin{bmatrix}81 & 27\\-1 & 4\end{bmatrix}$

*Ft incorrect/missing **U**$^{-1}$ for one product; ignore missing* $\frac{1}{13}$ *until the end*

A1

or $\frac{1}{13}\begin{bmatrix}108 & -1\\27 & 3\end{bmatrix}\begin{bmatrix}3 & 1\\-1 & 4\end{bmatrix}$

$= \begin{bmatrix}25 & 8\\6 & 3\end{bmatrix}$

CAO

A1

5

(c) **M**^{n} = **U D**^{n} **U**$^{-1}$

Including attempt to multiply

M1

D^n = $\begin{bmatrix}27^n & 0\\0 & 1\end{bmatrix}$

B1

M^n(1, 1) = $\frac{1}{13}\left(12 \times 27^n + 1\right)$

A1

So $4 \times 3 \times 3^{3n} + 1 = 4 \times 3^{3n+1} + 1$ div. by 13

Legitimately so from their working, from fact that the element is an integer

E1

Since **M** has all integer elements, each element of **M**^{n} is an integer also

Explaining why it must be an integer

E1

5

[16]

Q9.

(a) Det(**M**) = 1 $\Rightarrow$ Area invariant under T

2nd B1 ft ref. "area"

B1B1

2

(b) Char. Eqn. $\lambda^2 - 2\lambda + 1 = 0$

M1

$\Rightarrow \lambda = 1$ (twice)

A1

Substg. their λ back to find an evec:

M1

$\alpha\begin{bmatrix}2\\1\end{bmatrix}$

Any (non-zero) α

A1

(Since $\lambda = 1$) this represents a line of inv. pts.

ft if $\lambda \neq 1$

B1

5

(c) $\begin{bmatrix}-1 & 4\\-1 & 3\end{bmatrix}\begin{bmatrix}x\\\frac{1}{2}x+k\end{bmatrix}=\begin{bmatrix}x+4k\\\frac{1}{2}x+3k\end{bmatrix}$

M1A1

Verifying that $y' = \frac{1}{2}x' + k$

Be convinced AG

A1

3

[10]

Q10.

(a) (i) $\mathbf{M}^2 = \begin{bmatrix}2 & -1 & 1\\1 & 0 & 1\\1 & -1 & 2\end{bmatrix}\begin{bmatrix}2 & -1 & 1\\1 & 0 & 1\\1 & -1 & 2\end{bmatrix}$

M1

$$= \begin{bmatrix} 4 & -3 & 3 \\ 3 & -2 & 3 \\ 3 & -3 & 4 \end{bmatrix}$$

A1

$\mathbf{M}^2 + 2\mathbf{I}$ $= \begin{bmatrix} 4 & -3 & 3 \\ 3 & -2 & 3 \\ 3 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

$= \begin{bmatrix} 6 & -3 & 3 \\ 3 & 0 & 3 \\ 3 & -3 & 6 \end{bmatrix} = 3\mathbf{M}$

ie k = 3

A1 **3**

(ii) Multiplying by $\mathbf{M}^{-1}$

M1

to get $\mathbf{M} + 2\mathbf{M}^{-1} = 3\mathbf{I}$

ft

A1

so that $\mathbf{M}^{-1} = \frac{3}{2}\mathbf{I} - \frac{1}{2}\mathbf{M}$

ie $a = -\frac{1}{2}$ and $b = \frac{3}{2}$

A1 **3**

(b) (i) Char. eqn. is $\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$

One A mark for each of the other coefficients

M1A1A1A1

ie $(\lambda - 2)(\lambda - 1)^2 = 0$

Good factorisation attempt

M1

giving $\lambda_1 = 1$ (twice) and $\lambda_2 = 2$

A1 **6**

(ii) $\lambda = 1 \Rightarrow x - y + z = 0$ (thrice)

B1

Any two independent eigenvectors
Attempted

M1

(eg) $\alpha\begin{bmatrix}1\\1\\0\end{bmatrix}+\beta\begin{bmatrix}0\\1\\1\end{bmatrix}$

A1

$\lambda = 2 \Rightarrow -y + z = 0$
$x - 2y + z = 0 \Rightarrow x = y = z$
$x - y = 0$

M1

$\gamma\begin{bmatrix}1\\1\\1\end{bmatrix}$

A1 5

(iii) For $\lambda = 1$, eigenvectors represent a plane of invariant points
Plane

M1A1

For $\lambda = 2$, eigenvectors represent an invariant line

B1 3

[20]

Q11.

(a) $y = 0$ (or "x-axis") and $y = x$

or $\mathbf{r} = a\begin{bmatrix}1\\0\end{bmatrix}$, $\mathbf{r} = b\begin{bmatrix}1\\1\end{bmatrix}$

B1,B1

$y = 0$ is a line of invariant points since $\lambda = 1$
Allow if proven from $(x', y') = (x, y)$ or ft from their line corresponding to $\lambda = 1$

B1 3

(b) (i) $\mathbf{D} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, $\mathbf{U} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$,

*ft **U** from **D***

B1,B1

2

(ii) $\mathbf{U}^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

*ft from **U** (provided non-singular)*

B1

$\mathbf{M} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

Attempt

M1

$= \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ or $\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

ft first multiplication

A1

$= \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$

CAO *NMS* $\Rightarrow$ *0*

A1

4

(iii) $\mathbf{D}^n = \begin{bmatrix} 1 & 0 \\ 0 & 2^n \end{bmatrix}$

Noted or used

B1

$\mathbf{M}^n = \mathbf{U}\,\mathbf{D}^n\,\mathbf{U}^{-1}$

*Used; **must** actually do some multiplying*

M1

$= \begin{bmatrix} 1 & 2^n - 1 \\ 0 & 2^n \end{bmatrix}$

A1

3

[12]

Q12.

(a) For an *invariant line*, all points on the line have image points also on the line.

For a line of invariant points, all points on the line map onto themselves.

Clearly explained

B1

1

(b) Det = 3

B1

The s.f.. of area enlargement under T

Must mention area

B1

2

(c) Setting $x' = x$ and $y' = y$ and solving

$x = 2x - y$ and $y = -x + 2y$

M1

$\Rightarrow y = x$

A1

Or

Char. Eqn. is $\lambda^2 - 4\lambda + 3 = 0$

M1

$\lambda = 1$ gives l.o.i.p.s and $y = x$

A1

2

(d) (i) $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ -x + c \end{bmatrix}$

M1

$= \begin{bmatrix} 3x - c \\ -3x + 2c \end{bmatrix}$

A1

2

(ii) $y' = -x' + c$ also

M1

$\Rightarrow y = -x + c$ invariant for all c

A1

2

(e) Stretch, s.f. 3,

M1A1

perp^r. to $y = x$ (or || to $y = -x$)

A1

3

[12]

Q13.

(a) det **A** = 1

B1

1

(b) The z-axis (i.e. $x = y = 0$)

B1

1

(c) Rotation

M1

about the z-axis

ft axis or correct

A1ft

through $\cos^{-1} 0.28$

73.7° (awrt 74°) or 1.29 rads;

$\sin^{-1} 0.96$, $\tan^{-1} \frac{24}{7}$

A1

3

[5]

Q14.

(a) $\begin{vmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{vmatrix} = 67.84 - 51.84 = 16$

Attempt at det.

M1A1

2

(b) Inv. pts.g.b. $x' = x$, $y' = y$

B1

Subst[g]. in eqns. $6.4x - 7.2y = x$
$-7.2x + 10.6y = y$

M2
A1

$y = \frac{3}{4}x$

A1
5

Alt. I

Char. Eqn. is $\lambda^2 - 17\lambda + 16 = 0$

M1A1

$\lambda = 1$ or 16

A1

$\lambda = 1$ for l.o.i.p.s $\Rightarrow 5.4x - 7.2y = 0$

i.e. $y = \frac{3}{4}x$

ignore $\lambda = 16$ *work*

M1A1
(5)

Alt. II

$y = mx$ a l.o.i.p.s

$$\begin{bmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{bmatrix}\begin{bmatrix} x \\ mx \end{bmatrix} = \begin{bmatrix} 6.4x - 7.2mx \\ -7.2x + 10.6mx \end{bmatrix}$$

B1

$-7.2x + 10.6mx = m(6.4x - 7.2mx)$ also

M1
A1

$\Rightarrow 7.2m^2 + 4.2m - 7.2 = 0$

$\Rightarrow (4m - 3)(3m + 4) = 0$

A1

$\Rightarrow y = \frac{3}{4}$x or $y = -\frac{4}{3}x$

A1

Checking which one works

B1

(5)

[7]

EXAM PAPERS PRACTICE

Examiner reports

Q1.

The vast majority of students realised that $\mathbf{M}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$, and were able to multiply out to produce two equivalent equations. However, a significant number of these students were unable to use their equation(s) to find an invariant point. Many chose to solve them as simultaneous equations, and either made an error or confused themselves when every term cancelled out. A minority of students knew that (0, 0) would be invariant and gave this as the only possible point.

Q3.

The two transformations were often correctly described, although on occasions the plane $z = 0$ became 'an axis' or 'line'. A small minority of students reversed the order of multiplication of the two matrices. In part (c) an eigenvector approach rarely scored many marks due to full determinant expansion errors, incorrectly setting the determinant of $\mathbf{M}_3 = 0$ or not realising that 1 was an eigenvalue. The approach of $\mathbf{M}_3\mathbf{v} = \mathbf{v}$ was far more successful apart from minor manipulation errors.

Q4.

This question proved to be the most demanding. Many students knew what to do for part (a) but often failed to show all working out leading to a loss of marks. Part (b)(i) proved elusive for all but the most able students. When asked to find all invariant lines the most efficient and valid method involves obtaining y' and x' in terms of x, y and c and then trying to make them fit the equation $y' = mx' + c$. This method was rarely seen and yet is the correct standard approach. Other approaches such as using eigenvalues are more appropriate to lines of invariant points only.

Q5.

This question was also generally well done. The only commonly lost mark was that for part (b) which should have been very straightforward; perhaps too straightforward.

Q6.

This question produced many more at least partially successful attempts. Nonetheless, sign errors and arithmetical slips abounded and there was, yet again, very little evidence of tracking back to check obviously incorrect working and answers. It was only a very small minority who appreciated that a repeated eigenvalue should lead to an invariant plane, which needs two representative eigenvectors.

Uncertainty in parts (b) and (c) often led to marks not exceeding 10 out of the 15 available, even amongst the attempts of the better candidates.

Q7.

The structure of these questions proved very helpful to candidates, and those who were reasonably careful could generally score at least 16 of this question's 18 marks without any difficulty.

Q8.

This question was handled very well indeed in its technical aspects – writing down eigenvalues and eigenvectors; finding a 2 × 2 inverse matrix; determining **M**, and the accompanying matrix multiplications. So a lot of candidates scored lots of marks on this question. However, there were very few candidates indeed who managed to score more than 12 or 13, due to the explanations that were being looked for. At its simplest level, this began in part (a)(i), where candidates were required to write down "the eigenvalues and ***corresponding*** eigenvectors …" – in other words, to say which went with which. Examiners did not accept a fortuitously ordered separate pair of each (with a 50-50 chance of a correct pairing). In part (a)(ii), we also wanted to know **why** they had chosen one of the two possible lines as a line of invariant points (as opposed to just an invariant line).

Part (c) had been designed to require thought of the candidates, so the difficulties that arose here were not unexpected. There were two marks for explaining why the given expression was divisible by 13. The first was awarded to anyone who noted that the 1-1 element of $\mathbf{M}^n$ is an integer; the second, subtler, one was for the explanation of why it had to be an integer. Disappointingly few candidates got both marks.

Q9.

This question required several different ideas. In part (a), it was essential to use the word "area" to describe the significance of **det M** in relation to **T**, although responses were much more frequently appropriate in this respect than had been the case the last time such a question appeared. A similar type of comment was required at the end of part (b). In part (c), those who parametrised the line as $(x, \tfrac{1}{2}x + k)$ generally coped very well, although a significant number lost the final mark by not showing the necessary working to support the given answer carefully, as opposed to merely stating what had been stated in the question.

Q10.

This question was definitely the most demanding one on the paper; partly for its length, and partly due to working with a 3 × 3 matrix. On top of this, there was a repeated eigenvalue to the matrix, and a lot of candidates clearly did not know quite what to make of it. Nonetheless, even the relatively weaker candidates were still able to gain up to 12 of the marks.

Part (a)(i) was very well handled by those who attempted it. Part (a)(ii) was less well handled. Many of those who fell down here did so because they then went off to find the inverse matrix $\mathbf{M}^{-1}$ itself, rather than follow the logic of the question. Many more who did multiply the matrix equation by $\mathbf{M}^{-1}$ got tangled up in some way – failing to spot that $\mathbf{M}\,\mathbf{M}^{-1} = \mathbf{I}$, for instance, or being unable to divide by 2 under exam conditions.

In part (b), the factorisation of the determinant $|\mathbf{M} - \lambda\mathbf{I}|$ was usually approached by row/column operations, which was very pleasing to see. However, it frequently (actually usually) led to the strange sight of candidates taking a partially or completely factorised expression, and then multiplying it out to get a standard cubic form, only to start to factorise it again to find the three eigenvalues.

Those who made a mistake in this working then shot themselves in the foot, losing the following method mark for the attempt to factorise, which they failed to pick up by presuming that their cubic actually *was* equal to the factorised form $(\lambda - 1)^2(\lambda - 2)$ rather than doing any of the necessary working or going back to check their previous, incorrect working.

The final two parts to the question were demanding, but actually not as difficult as candidates found them to be. Despite the fact that most candidates getting this far quite correctly deduced that $x - y + z = 0$ for the repeated eigenvalue $\lambda = 1$, almost all of them then believed this represented the equation of a line. This error then prevented them from finding a pair of independent representative eigenvectors (or an alternative parametric plane form) and also from describing the result as a plane of invariant points, rather than as a line.

Q11.

Most difficulties with this question arose in part (a), where candidates had to consider the meaning and relevance of the information given to them in the question. The most commonly lost mark was in the widespread failure to point out that the line of invariant points was flagged up by the fact that the eigenvalue corresponding to the x-axis was 1. In general, most other marks lost on this question arose as a result of minor slips and oversights in the working. A few candidates didn't seem to know how to find $\mathbf{M}^n$.

Q12.

In part (b), the determinant of the transformation matrix is the scale factor of area enlargement under T – very few candidates used the key word "area" here. In part (d), having found the image of $(x, -x + c)$ to be $(x', y') = (3x - c, 2c - 3x)$, which many successfully did, candidates were supposed simply to show that $y' = -x' + c$. Few candidates understood that this was what was required.

Traditionally, describing transformations is very poorly done, and this was no exception. There was a lack of understanding as to how the algebra, matrix algebra and coordinate geometry all tie together with the geometry of transformations, and some more practice was needed for most candidates on this score.

Q13.

Most candidates answered this question very well, making intelligent use of the results given in the formula booklet. Once again, however, there was a significant proportion of candidates who gave an inappropriate answer, often $z = 0$, when asked for the invariant line in part (b).

Q14.

Candidates found this question difficult. The answer to part (a) is the determinant of the given matrix; it was not uncommon for candidates to find this and then take its square root and offer this as the scale factor required. Although there are at least three valid approaches to part (b), most candidates' uncertainty led to a mish-mash of algebra, with the correct answer often embedded in so much other working that the candidate producing it seemed to have no idea that they had just done the necessary work.